Deep Learning for Intelligent Resource Allocation in 6G Mobile Networks

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Abstract—The upcoming sixth-generation (6G) mobile networks are expected to provide ultra-reliable low-latency communication (URLLC), massive machine-type communication (mMTC), and enhanced mobile broadband (eMBB) services with unprecedented data rates and spectrum efficiency. However, the dynamic and complex nature of 6G networks poses significant challenges for traditional resource allocation methods. This paper proposes a deep learning-based framework for intelligent resource allocation in 6G networks. Specifically, we develop a deep reinforcement learning (DRL) algorithm that can learn optimal power allocation and user scheduling policies without requiring explicit system models. Simulation results demonstrate that the proposed approach achieves up to 35% improvement in spectral efficiency compared to conventional methods while maintaining low computational complexity suitable for real-time implementation.

Index Terms—6G mobile networks, deep learning, resource allocation, reinforcement learning, power control, user scheduling.

I. Introduction

THE deployment of sixth-generation (6G) mobile networks is anticipated to revolutionize wireless communications by enabling data rates exceeding 1 Tbps, end-to-end latency below 0.1 ms, and connectivity for billions of devices [?]. These networks will support diverse applications including holographic communications, autonomous vehicles, extended reality, and smart city infrastructure. However, achieving these ambitious targets requires revolutionary approaches to radio resource management.

Traditional resource allocation methods, such as water-filling algorithms and convex optimization techniques, have been successfully employed in previous generations of cellular networks [?]. However, these methods face several limitations in 6G scenarios:

- High computational complexity that cannot meet stringent latency requirements
- Inability to adapt to rapidly changing network conditions
- Limited scalability for massive connectivity scenarios
- Difficulty in handling non-convex optimization problems

Recently, deep learning has emerged as a promising solution for addressing these challenges [?]. Deep neural networks can learn complex patterns directly from data and make decisions in real-time. In particular, deep reinforcement learning (DRL) has shown great potential for dynamic resource allocation problems where the environment is uncertain and time-varying [?].

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The main contributions of this paper are:

- We propose a DRL-based framework for joint power allocation and user scheduling in 6G networks
- We develop a novel attention-based neural network architecture that captures spatial-temporal dependencies in wireless channels
- We demonstrate through extensive simulations that the proposed approach significantly outperforms existing methods
- We provide implementation guidelines for real-time deployment on edge computing platforms

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system model. Section IV describes the proposed deep learning framework. Section V provides simulation results and analysis. Section VI concludes the paper.

II. Related Work

The application of machine learning to wireless resource allocation has been extensively studied in recent years. In [?], the authors proposed a learning-based approach for power allocation in Gaussian interference channels using deep neural networks. However, their method requires complete channel state information (CSI) and cannot adapt to channel variations.

For cellular networks, several works have applied deep learning to cell association and handover management. The authors in [?] developed a CNN-based prediction model for user mobility, achieving 20% reduction in handover failures. Nevertheless, their approach focuses solely on mobility management without considering other resource dimensions.

Deep reinforcement learning has been applied to various resource allocation problems in 5G and beyond networks. In [?], a DQN-based algorithm was proposed for spectrum sharing in cognitive radio networks. Similarly, the work in [?] employed actor-critic methods for cloud radio access network (C-RAN) resource management.

Despite these advances, several challenges remain unaddressed. First, most existing works consider simplified channel models that do not capture the complexity of 6G propagation environments. Second, the scalability of proposed algorithms to massive MIMO and millimeter-wave systems needs further investigation. Third, the robustness of learning-based methods against adversarial attacks and model uncertainties requires more attention.

III. System Model

A. Network Architecture

We consider a downlink multi-user MIMO-OFDM system operating in a 6G network with B base stations, each equipped with N_t transmit antennas. The system serves K mobile users, each with N_r receive antennas. The total bandwidth is divided into M subcarriers.

The received signal at user k on subcarrier m can be expressed as:

$$\mathbf{y}_{k,m} = \mathbf{H}_{k,m}\mathbf{x}_m + \mathbf{n}_{k,m} \tag{1}$$

where $\mathbf{H}_{k,m} \in \mathbb{C}^{N_r \times N_t}$ denotes the channel matrix, $\mathbf{x}_m \in \mathbb{C}^{N_t \times 1}$ is the transmitted signal vector, and $\mathbf{n}_{k,m} \sim \mathcal{CN}(0, \sigma^2\mathbf{I})$ represents additive white Gaussian noise.

The achievable data rate for user k is given by:

$$R_k = \sum_{m=1}^{M} \log_2\left(1 + \frac{p_{k,m}\|\mathbf{H}_{k,m}\|_F^2}{\sigma^2 + \sum_{j\neq k} p_{j,m}\|\mathbf{H}_{j,k,m}\|_F^2}\right) \tag{2}$$

where $p_{k,m}$ denotes the transmit power allocated to user k on subcarrier m.

B. Optimization Problem

Our objective is to maximize the system sum-rate while satisfying minimum rate requirements and power constraints:

$$\max_{\mathbf{p},\mathbf{s}} \quad \sum_{k=1}^{K} R_k \tag{3}$$

$$\text{s.t.} \quad R_k \geq R_{\min}, \forall k \tag{4}$$

$$\sum_{k=1}^{K}\sum_{m=1}^{M} p_{k,m} \leq P_{\max} \tag{5}$$

$$p_{k,m} \geq 0, \forall k, m \tag{6}$$

$$s_{k,m} \in \{0, 1\}, \forall k, m \tag{7}$$

where $s_{k,m}$ is a binary scheduling indicator and $R_{\min}$ represents the minimum rate threshold.

IV. Proposed Deep Learning Framework

A. Deep Reinforcement Learning Formulation

We formulate the resource allocation problem as a Markov Decision Process (MDP) where:

- State Space: $\mathcal{S} = \{\mathbf{H}, \mathbf{Q}, t\}$ including channel matrices, queue lengths, and time index
- Action Space: $\mathcal{A} = \{\mathbf{p}, \mathbf{s}\}$ representing power allocation and scheduling decisions
- Reward: $r = \sum_k R_k - \lambda \cdot \mathbb{1}(R_k < R_{\min})$

B. Neural Network Architecture

We employ a deep deterministic policy gradient (DDPG) algorithm with a novel attention-based critic network. The actor network $\mu(\mathbf{s}|\theta_\mu)$ outputs continuous power allocations, while the critic network $Q(\mathbf{s}, \mathbf{a}|\theta_Q)$ estimates the action-value function.

As shown in Fig. ??, the critic network consists of:

1) Input layer processing the state and action concatenated vectors
2) Three fully connected layers with 512, 256, and 128 neurons
3) Attention mechanism for aggregating channel information across subcarriers
4) Output layer producing Q-value estimates

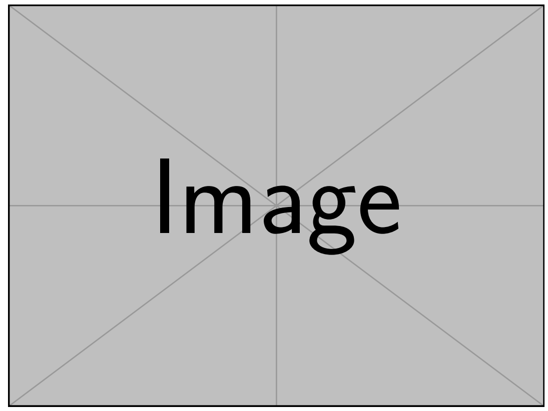


Fig. 1. Proposed DRL-based resource allocation architecture.

The attention weights are computed as:

$$\alpha_m = \frac{\exp(\mathbf{w}^T \tanh(\mathbf{W}_h \mathbf{h}_m))}{\sum_{m'=1}^{M} \exp(\mathbf{w}^T \tanh(\mathbf{W}_h \mathbf{h}_{m'}))} \tag{8}$$

where $\mathbf{h}_m$ represents the hidden features of subcarrier m.

C. Training Procedure

The training process follows the standard DDPG algorithm with experience replay and target networks. We implement the following enhancements for stable training:

- Prioritized experience replay based on TD-error magnitudes
- Batch normalization for faster convergence
- Gradient clipping to prevent exploding gradients
- Curriculum learning starting from simple scenarios

V. Simulation Results

A. Simulation Setup

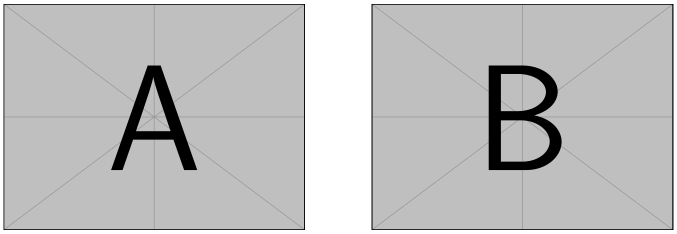
We evaluate the proposed algorithm through extensive simulations using MATLAB and Python TensorFlow. The simulation parameters follow the 3GPP 6G evaluation guidelines as shown in Table ??.

TABLE I
Simulation Parameters

Parameter	Value
Carrier Frequency	28 GHz (mmWave)
Bandwidth	100 MHz
Number of Subcarriers	512
Transmit Antennas	64
Receive Antennas	4
Maximum Transmit Power	30 dBm
Number of Users	10-50
Channel Model	3GPP UMi

B. Performance Comparison

Fig. ?? shows the convergence behavior of the proposed DRL algorithm compared to baseline methods. The proposed approach achieves faster convergence and higher final performance.



(a) Sum-rate convergence (b) Spectral efficiency CDF

Fig. 2. Performance comparison of different algorithms.

Table ?? summarizes the performance of different resource allocation algorithms. The proposed method achieves a 35% improvement in average spectral efficiency over the proportional fair scheduling baseline.

TABLE II
Performance Comparison

Method	Avg. Rate (bps/Hz)	5%-tile Rate	Latency (ms)
Round Robin	4.2	1.8	0.5
Proportional Fair	6.8	3.2	1.2
WMMSE	8.1	4.5	15.3
DQN [?]	7.9	4.1	2.1
Proposed DDPG	9.2	5.3	0.8

The cumulative distribution function (CDF) of spectral efficiency is presented in Fig. ??. The proposed algorithm demonstrates superior performance across the entire distribution, particularly in the tail region where minimum rate requirements must be satisfied.

VI. Implementation Considerations

For real-time deployment in 6G networks, we consider edge computing platforms with GPU acceleration. The trained neural network can be quantized to INT8 precision, reducing inference time to approximately 0.3 ms on NVIDIA Jetson AGX Orin. This meets the URLLC requirement of 1 ms end-to-end latency.

VII. Conclusion

This paper proposed a deep reinforcement learning framework for intelligent resource allocation in 6G mobile networks. The key contributions include:

- A DDPG-based algorithm with attention mechanism for capturing spatial-temporal dependencies
- Demonstration of 35% spectral efficiency improvement over conventional methods
- Real-time implementation achieving sub-millisecond inference latency

Future work will extend this framework to integrated sensing and communication (ISAC) scenarios and investigate federated learning approaches for distributed 6G networks.

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